🎯 Lecture 20: Practice Problems — Technical Case Studies

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📌 Case Study 1: High-Frequency FinTech Bayesian Fraud & Anomaly Detection Pipeline

🏢 Context & Engineering Problem

A tier-1 payment processor handles $10,000,000$ transactions daily. True fraudulent transactions are rare, with an established historical base rate.

1. State of Nature & Prior Base Rate

The transaction sample space Ω is partitioned into two mutually exclusive and exhaustive states of nature:

- F: Transaction is **Fraudulent**
- $L = F^c$: Transaction is **Legitimate**

Partition Invariants:

$$F \cap L = \emptyset, \qquad F \cup L = \Omega, \qquad P(F) + P(L) = 1$$

Established prior base rate:

$$P(F) = 0.002 \quad (0.2$$

2. Diagnostic Heuristic Signals (Likelihoods)

The risk engineering team deploys two sequential fraud detection heuristic signals:

1. **Signal 1 (S_1 — Device Fingerprint Anomaly):**
   - Sensitivity (True Positive Rate):

$$P(S_1 \mid F) = 0.85$$

- False Positive Rate:

$$P(S_1 \mid L) = 0.04$$

- Specificity (True Negative Rate):

$$P(S_1^c \mid L) = 1 - P(S_1 \mid L) = 0.96$$

2. **Signal 2 (S_2 — Impossible Geo-Velocity Movement):**
   - Sensitivity (True Positive Rate):

$$P(S_2 \mid F) = 0.90$$

- False Positive Rate:

$$P(S_2 \mid L) = 0.03$$

- Specificity (True Negative Rate):

$$P(S_2^c \mid L) = 1 - P(S_2 \mid L) = 0.97$$

3. Conditional Independence Axiom

Assume that conditional on transaction class (F or L), signals S_1 and S_2 are conditionally independent:

$$P(S_1 \cap S_2 \mid F) = P(S_1 \mid F) \cdot P(S_2 \mid F)$$

$$P(S_1 \cap S_2 \mid L) = P(S_1 \mid L) \cdot P(S_2 \mid L)$$

🎯 Mathematical & Implementation Tasks

Task 1: Single-Signal Bayesian Update (S_1)

1. **Law of Total Probability:** Express and compute the total marginal evidence $P(S_1)$:
$$P(S_1) = P(S_1 \mid F)P(F) + P(S_1 \mid L)P(L)$$
2. **Bayes' Theorem Formulation:** Derive and compute the posterior probability of fraud given Signal 1:
$$P(F \mid S_1) = \frac{P(S_1 \mid F)P(F)}{P(S_1)} = \frac{P(S_1 \mid F)P(F)}{P(S_1 \mid F)P(F) + P(S_1 \mid L)P(L)}$$
3. **Base Rate Fallacy Analysis:** Explain mathematically why $P(F \mid S_1) \ll P(S_1 \mid F)$ despite high test sensitivity when $P(F) \ll P(L)$.

Task 2: Sequential Bayesian Updating (Recursive Bayes)

1. Treat the posterior from Task 1 as the updated prior belief:
$$\pi_1(F) = P(F \mid S_1), \qquad \pi_1(L) = P(L \mid S_1) = 1 - \pi_1(F)$$
2. Compute marginal evidence for Signal 2 conditional on Signal 1:
$$P(S_2 \mid S_1) = P(S_2 \mid F)\pi_1(F) + P(S_2 \mid L)\pi_1(L)$$
3. Derive the sequentially updated posterior belief:
$$P(F \mid S_1, S_2) = \frac{P(S_2 \mid F)\pi_1(F)}{P(S_2 \mid S_1)} = \frac{P(S_2 \mid F)P(F \mid S_1)}{P(S_2 \mid F)P(F \mid S_1) + P(S_2 \mid L)P(L \mid S_1)}$$

Task 3: Joint Simultaneous Likelihood Verification

Prove mathematically that sequential updating yields the exact same posterior as the joint simultaneous likelihood formulation:

$$P(F \mid S_1 \cap S_2) = \frac{P(S_1 \cap S_2 \mid F)P(F)}{P(S_1 \cap S_2)} = \frac{P(S_1 \mid F)P(S_2 \mid F)P(F)}{P(S_1 \mid F)P(S_2 \mid F)P(F) + P(S_1 \mid L)P(S_2 \mid L)P(L)}$$

Task 4: Odds & Bayes Factor Formulation

Express the update using Bayes' Rule in Odds form:

$$\text{Prior Odds:} \quad O(F) = \frac{P(F)}{P(L)}$$

$$\text{Bayes Factor (Likelihood Ratio):} \quad \Lambda_1 = \frac{P(S_1 \mid F)}{P(S_1 \mid L)}, \qquad \Lambda_2 = \frac{P(S_2 \mid F)}{P(S_2 \mid L)}$$

$$\text{Posterior Odds:} \quad O(F \mid S_1, S_2) = O(F) \cdot \Lambda_1 \cdot \Lambda_2$$

$$\text{Posterior Probability Recovery:} \quad P(F \mid S_1, S_2) = \frac{O(F \mid S_1, S_2)}{1 + O(F \mid S_1, S_2)}$$

📌 Case Study 2: Bioinformatics Clinical Drug Efficacy via Binomial Testing

🏢 Context & Engineering Problem

A biotechnology firm is conducting a Phase II clinical oncology trial for a novel immunotherapy agent across $n = 50$ terminal cancer patients.

- **Historical Standard-of-Care (SOC) Remission Rate:** $p_0 = 0.20$ (20 historical response rate).
- **Trial Outcome:** At the conclusion of the trial, $k = 18$ patients achieve verified complete tumor remission.

1. Random Variable & Probability Model

Let $X = \sum_{i=1}^{n} Y_i$ denote the total number of remissions, where each patient outcome $Y_i \sim \text{Bernoulli}(p)$:

$$X \sim \text{Binomial}(n, p), \qquad X \in 0, 1, 2, \ldots, n$$

Binomial Probability Mass Function (PMF):

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

Expectation and Variance:

$$\mathbb{E}[X] = np, \qquad \text{Var}(X) = np(1-p), \qquad \sigma = \sqrt{np(1-p)}$$

🎯 Mathematical & Implementation Tasks

Task 1: Hypothesis Formulation

Formulate the formal statistical hypotheses for a one-tailed efficacy trial:

$$\begin{cases} H_0 : p \leq p_0 = 0.20 \quad (\text{Null Hypothesis: New drug is no better than SOC}) \\ H_1 : p > p_0 = 0.20 \quad (\text{Alternative Hypothes} \end{cases}$$

Task 2: Exact Binomial Tail Probability (p-value)

Under null hypothesis $H_0 : X \sim B(50, 0.20)$, derive and compute the exact one-tailed p-value:

$$p\text{-value} = P(X \geq 18 \mid n = 50, p_0 = 0.20) = \sum_{j=18}^{50} \binom{50}{j} (0.20)^j (0.80)^{50-j}$$

Complementary Cumulative Distribution Function (CDF) form:

$$p\text{-value} = 1 - P(X \leq 17 \mid H_0) = 1 - \sum_{j=0}^{17} \binom{50}{j} (0.20)^j (0.80)^{50-j}$$

Task 3: Gaussian (Normal) Approximation via Central Limit Theorem

1. Compute the expected mean μ and standard deviation σ:

$$\mu = np_0 = 50 \times 0.20 = 10$$

$$\sigma = \sqrt{np_0(1-p_0)} = \sqrt{50 \times 0.20 \times 0.80} = \sqrt{8} \approx 2.8284$$

2. Verify the Rule-of-Thumb validity condition:

$$np_0 = 10 \geq 5 \quad \text{and} \quad n(1-p_0) = 40 \geq 5$$

3. Using Yates' continuity correction ($k - 0.5 = 17.5$), compute the standardized Z-score:

$$Z = \frac{(k - 0.5) - \mu}{\sigma} = \frac{(18 - 0.5) - 10}{\sqrt{8}} = \frac{7.5}{2.8284} \approx 2.6516$$

4. Evaluate the Gaussian p-value using the Standard Normal CDF $\Phi(z)$:

$$p_{\text{approx}} = P(Z \geq 2.6516) = 1 - \Phi(2.6516) = 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{2.6516} e^{-\frac{u^2}{2}} du$$

Task 4: Statistical Decision Rule & Regulatory Inference

Given standard clinical significance threshold $\alpha = 0.01$ (99% confidence level):

$$\text{Decision Rule:} \quad \text{Reject } H_0 \iff p\text{-value} < \alpha$$

Assess percentage discrepancy between the exact Binomial and Gaussian approximation results, and deliver the formal Phase III approval recommendation.

📌 Case Study 3: Autonomous Vehicle LiDAR Sensor Calibration & Gaussian Noise Normalization

🏢 Context & Engineering Problem

An autonomous vehicle navigation subsystem uses a pulsed LiDAR sensor to estimate distance to roadside obstacles.

1. Stochastic Error Model

Distance measurement model:

$$X_{\text{meas}} = d_{\text{true}} + \epsilon$$

Where measurement error ϵ is an additive zero-mean continuous Gaussian random variable:

$$\epsilon \sim \mathcal{N}(\mu, \sigma^2), \qquad \mu = 0, \quad \sigma = 0.15 \text{ meters}$$

Probability Density Function (PDF):

$$f(\epsilon; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\epsilon - \mu)^2}{2\sigma^2}\right) = \frac{1}{0.15\sqrt{2\pi}} \exp\left(-\frac{\epsilon^2}{2(0.15)^2}\right)$$

Standard Normal CDF:

$$\Phi(z) = P(Z \leq z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-\frac{u^2}{2}} du = \frac{1}{2}\left[1 + \text{erf}\left(\frac{z}{\sqrt{2}}\right)\right]$$

🎯 Mathematical & Implementation Tasks

Task 1: 68-95-99.7 Empirical Rule Zone Probabilities

Evaluate the exact theoretical probabilities of the three operational safety zones:

1. **Green Zone (Normal Operation: $|\epsilon| \leq 1\sigma$):**

$$P(|\epsilon| \leq \sigma) = P(-\sigma \leq \epsilon \leq \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\sigma}^{\sigma} e^{-\frac{\epsilon^2}{2\sigma^2}} d\epsilon$$

$$P(|\epsilon| \leq 0.15) = \Phi(1) - \Phi(-1) = 2\Phi(1) - 1 \approx 0.68269 \quad (68.27$$

2. **Yellow Zone (Advisory Warning: $1\sigma < |\epsilon| \leq 2\sigma$):**

$$P(\sigma < |\epsilon| \leq 2\sigma) = 2 \cdot [\Phi(2) - \Phi(1)] \approx 2 \cdot (0.97725 - 0.84134) = 0.27181 \quad (27.18$$

3. **Red Zone (Critical Sensor Hazard: $|\epsilon| > 3\sigma$):**

$$P(|\epsilon| > 3\sigma) = 2 \cdot [1 - \Phi(3)] \approx 2 \cdot (1 - 0.99865) = 0.00270 \quad (0.27$$

Task 2: Z-Score Standardization & Tail Probability Integration

Derive the probability that measurement error exceeds safety cutoff $\delta = 0.25$ meters:

1. Standardized Cutoff Z-Score:

$$Z_{\text{cutoff}} = \frac{\delta - \mu}{\sigma} = \frac{0.25 - 0}{0.15} = \frac{5}{3} \approx 1.6667$$

2. Two-Tailed Exceedance Probability:

$$P(|\epsilon| > 0.25) = P(\epsilon > 0.25) + P(\epsilon < -0.25) = 2 \cdot P(Z > 1.6667)$$

$$P(|\epsilon| > 0.25) = 2 \cdot \left[1 - \Phi\left(\frac{0.25}{0.15}\right)\right] = 2 \cdot \left[1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{1.6667} e^{-\frac{u^2}{2}} du\right]$$

$$P(|\epsilon| > 0.25) = 2 \cdot (1 - 0.95221) \approx 0.09558 \quad (9.56$$

Task 3: Monte Carlo Calibration Simulation

Simulate $N = 1,000,000$ synthetic LiDAR sensor pings to verify empirical convergence:

1. **Sample Mean Convergence (Weak Law of Large Numbers):**

$$ \bar{\epsilon}_N = \frac{1}{N} \sum_{i=1}^N \epsilon_i \xrightarrow{P} \mathbb{E}[\epsilon] = 0 $$

2. **Sample Variance Unbiased Estimator:**

$$s_N^2 = \frac{1}{N-1} \sum_{i=1}^N (\epsilon_i - \bar{\epsilon}_N)^2 \xrightarrow{P} \sigma^2 = 0.15^2 = 0.0225$$

3. **Empirical Critical Violation Rate:**

$$ \hat{p}_{\text{fail}} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}\left(|\epsilon_i| > 0.25 \right) \xrightarrow{P} P(|\epsilon| > 0.25) \approx 0.0956 $$